Pankaj Yadav

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linkedin.com/in/pankaj-yadav-867a40200 | github.com/rogue816/Experiments
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Research Interests

Learning from imbalanced and rare-event data; certified and theoretically grounded oversampling; generative models (GANs, VAEs, diffusion) for data augmentation; cost-aware decision thresholds and calibration under prevalence shift; interpretable machine learning for healthcare (geriatrics, sarcopenia, fall risk); tail-risk modelling in climate finance.

Education

Ph.D., Mathematics Jan 2022 – Dec 2026 (expected)
Indian Institute of Technology Jodhpur, India
Thesis: *Rare Regions in Learning and Decision-Making: Certified Oversampling, Local Support, and Thresholds*
Supervisor: Dr. V. V. M. S. Chandramouli CGPA: 8.7/10

M.Sc., Mathematics 2019 – 2021
Indian Institute of Technology Jodhpur, India
CGPA: 7.36/10

B.Sc. (Hons.), Mathematics 2015 – 2018
University of Delhi, New Delhi, India
CGPA: 6.662/10

Doctoral Research

The thesis studies *rare regions*, sparse minority classes and distribution tails, and how they affect learning and downstream decisions. It is organised in four parts:

- **Generation — CISO.** An interpolation oversampler with the first per-instance certified clearance guarantee in its family, plus a provably monotone safety parameter. Evaluated on 45 datasets $\times$ 4 classifiers $\times$ 11 methods, it completed all 39,900 fold-level runs without failure (a clustering baseline failed on 50.3%), while matching SMOTE on PR-AUC.
- **Diagnosis — KANs under imbalance.** A parameter-matched comparison of Kolmogorov–Arnold Networks, MLPs and XGBoost on 10 datasets. KANs move from last (mean rank 8.40) to first (1.70) under resampling. Inspection of learned splines explains why.
- **Admissibility — SAvRAC.** A cost-aware threshold framework for three clinical care settings, with a dataset suitability score that tests whether data can support a clinically admissible decision rule at all.
- **Transfer — rainfall hedging.** Extends the same logic to distribution tails: a quantile–gamma XGBoost/SARIMA framework for weather-derivative triggers, with reliability reported separately for drought and flood regimes.

Awards and Fellowships

Best Student Paper Award, ICT4AWE 2022 (8th Int. Conf. on ICT for Ageing Well and e-Health) 2022
Junior Research Fellowship (JRF), CSIR 2019
CSIR–NET, Mathematical Sciences — All India Rank 158 2019
IIT–JAM, Mathematics — All India Rank 598 2019

Publications

Journal Articles

[J1] **P. Yadav** and V. Vijay. Hedging rainfall volatility in Indian agriculture: A quantile-gamma approach to weather derivatives. *Journal of Climate Finance*, vol. 14, p. 100082, 2026. doi:10.1016/j.jclimf.2026.100082

[J2] **P. Yadav** and V. Vijay. Are Kolmogorov–Arnold Networks (KANs) suitable for imbalanced data classification? *Accepted, IEEE Transactions on Emerging Topics in Computational Intelligence*.

Conference Papers and Book Chapters

[C1] G. Sihag, **P. Yadav**, V. Delcroix, V. Vijay, X. Siebert, S. Yadav, and F. Puisieux. Evaluation of risk factors for fall in elderly people from imbalanced data using the oversampling technique SMOTE. *Proc. ICT4AWE 2022*, 2022. doi:10.5220/0011041200003188 **(Best Student Paper Award)**

[C2] G. Sihag et al. (incl. **P. Yadav**). Advantages of oversampling techniques: A case study in risk factors for fall prediction. In *ICT for Ageing Well and e-Health (ICT4AWE 2021/2022)*, CCIS vol. 1856, Springer, 2023. doi:10.1007/978-3-031-37496-8_4

[C3] **P. Yadav**, G. Sihag, V. P. Singh, and V. Vijay. Solar PV power generation forecasting using synergy of artificial intelligence and metaheuristics based approaches. *IACC*, pp. 210–225, Springer, 2025. doi:10.1007/978-3-031-84602-1_15

[C4] V. Vijay, B. K. Shukla, S. K. Yadav, and **P. Yadav**. Prediction of muscle power in elderly using functional screening data. *IEEE MIPR*, 2023. doi:10.1109/MIPR59079.2023.00017

Under Review / Submitted

[R1] **P. Yadav** and V. Vijay. Certified Interpolation Safe Oversampling (CISO): Per-instance safety guarantees for imbalanced learning. Submitted. arXiv:2501.15790

[R2] **P. Yadav**, G. Sihag, and V. Vijay. Generative adversarial networks (GANs) for handling imbalance in data mining: Methods, taxonomies, and challenges. Under review. arXiv:2502.16535

[R3] **P. Yadav** and V. Vijay. Population-calibrated risk learning under prevalence shift: A sarcopenia case study. Under review, 2026.

[R4] **P. Yadav**, B. K. Shukla, and J. Bassement. SAvRAC: A clinically constrained and interpretable framework for sarcopenia assessment. Under review.

[R5] **P. Yadav**, R. K. Singh, and V. Vijay. Interpretable trading decision support across market microstructures: Adaptive fuzzy logic versus stochastic and information-theoretic systems. Under review, 2026.

Conference Presentations

ICT4AWE 2022 — Evaluation of risk factors for fall in elderly people from imbalanced data using SMOTE (Best Student Paper) 2022

IACC 2025 — Solar PV power generation forecasting using AI and metaheuristics 2025

Teaching Experience

Teaching Assistant, Department of Mathematics, IIT Jodhpur (2023 – 2025), for B.Tech, M.Tech, Executive M.Tech and PG Diploma programmes:

- **Probability, Statistics and Stochastic Processes** (B.Tech) — Jul–Dec 2023, Aug–Dec 2024, Aug–Nov 2025
- **Linear Algebra and Differential Equations** (B.Tech) — Jan–Jul 2023, Jan–May 2025
- **Financial Engineering** (B.Tech/M.Tech, Jan–May 2024; Executive M.Tech, Jan–May 2025)
- **Financial Analytics** (PG Diploma in Data Science) — May–Jul 2024, Oct–Dec 2024
- **Statistical Techniques** — Aug–Nov 2025
- **Applied Mathematics and Statistics** (DE) — Jul–Aug 2025

Technical Skills

Programming	Python, R, MATLAB
ML / DL	PyTorch, TensorFlow, scikit-learn, XGBoost
Methods	Imbalanced learning, oversampling (SMOTE variants, GANs, VAEs, diffusion), KANs, cost-sensitive thresholding, probability calibration, quantile regression, SARIMA, non-parametric statistical testing (Friedman, Wilcoxon)
Tools	LaTeX, Git

Leadership and Service

General Secretary, Hostel B3 Sep 2024 – Oct 2025
IIT Jodhpur

Head, Institute Dining Committee Sep 2024 – Sep 2025
IIT Jodhpur

References

Dr. V. V. M. S. Chandramouli
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Dr. Sandeep Kumar Yadav
Professor, Department of Electrical Engineering
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